

Appendix A

Covariance matrix

Assoc. Prof. Dr. Duong Tuan Anh
Faculty of Computer Science and Engineering,
HCMC Univ. of Technology
9/2015

How to compute covariance matrix

- Given two attributes X_i, X_j with the two sets of values:

$$X_i = \{x_{i1}, x_{i2}, \dots, x_{in}\}, X_j = \{x_{j1}, x_{j2}, \dots, x_{jn}\}$$

(n : the number of samples)

- Covariance between two attributes X_i, X_j :

$$\begin{aligned}\sigma_{ij} &= \text{cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)] \\ &= [(x_{i1} - \mu_i)(x_{j1} - \mu_j) + (x_{i2} - \mu_i)(x_{j2} - \mu_j) + \dots + (x_{in} - \mu_i)(x_{jn} - \mu_j)]/n \\ &= [a_{i1} \cdot a_{j1} + a_{i2} \cdot a_{j2} + \dots + a_{in} \cdot a_{jn}]/n\end{aligned}$$

$$= \frac{1}{n} \sum_{k=1}^n a_{ik} a_{jk}$$

$$= \frac{1}{n} A_i A_j^T$$

where A_i and A_j are X_i and X_j after subtracting off the means μ_i and μ_j , respectively

- For convenience, assume that X_i and X_j had been subtracted off its corresponding means.
- Now we can generalize from two attributes to an arbitrary number of attributes. Define a new $m \times n$ matrix $\mathbf{X}$.

$$\mathbf{X} = \begin{bmatrix} x_1 \\ \cdot \\ \cdot \\ x_m \end{bmatrix}$$

Each *row* in $\mathbf{X}$ corresponds to all values of a particular attribute. Each *column* of $\mathbf{X}$ corresponds to a set of attribute values from a particular sample.

Now we arrive at a definition for the covariance matrix $\mathbf{C}_\mathbf{X}$.

$$\mathbf{C}_\mathbf{X} = (1/n)\mathbf{X}\mathbf{X}^T$$

- Consider the matrix $\mathbf{C}_X = (1/n)\mathbf{X}\mathbf{X}^T$. The ij -th element of $\mathbf{C}_X$ is the dot product between the vector of the i -th attribute with the vector of the j -th attribute.
- We can summarize several properties of $\mathbf{C}_X$:
 - $\mathbf{C}_X$ is a square symmetric $m \times m$ matrix.
 - The diagonal terms of $\mathbf{C}_X$ are the variances of particular attributes.
 - The off-diagonal terms of $\mathbf{C}_X$ are the covariances between attributes.
- Note: In practice, the covariance σ_{AB} is calculated as

$$\frac{1}{n-1} \sum_{i=1}^n a_i b_i$$

The slight change in normalization constant arises from estimation theory.

Example

- Five measurements (observations) each are taken of 3 features, X_1 , X_2 and X_3 so that the feature vector is

X_1	X_2	X_3
4.0	2.0	0.6
4.2	2.1	0.59
3.9	2.0	0.58
4.3	2.1	0.62
4.1	2.2	0.63

The mean values are given by $\mu = [4.1 \quad 2.08 \quad 0.604]$.

The matrix $\mathbf{X}$ can be written as:

$$\begin{bmatrix} 4.0 & 4.2 & 3.9 & 4.3 & 4.1 \\ 2.0 & 2.1 & 2.0 & 2.1 & 2.2 \\ 0.6 & 0.59 & 0.58 & 0.62 & 0.63 \end{bmatrix}$$

After subtracting off the mean, the matrix $\mathbf{X}$ become:

$$X = \begin{bmatrix} -0.1 & 0.1 & -0.2 & 0.2 & 0 \\ -0.08 & 0.02 & -0.08 & 0.02 & 0.12 \\ -0.004 & -0.014 & -0.024 & 0.016 & 0.026 \end{bmatrix}$$

And the transposed matrix $\mathbf{X}^T$:

$$X^T = \begin{bmatrix} -0.1 & -0.08 & -0.004 \\ 0.1 & 0.02 & -0.014 \\ -0.2 & -0.08 & -0.024 \\ 0.2 & 0.02 & 0.016 \\ 0 & 0.12 & 0.026 \end{bmatrix}$$

Finally, the covariance matrix: $\Sigma = (1/4)\mathbf{XX}^T$

$$\Sigma = \begin{bmatrix} 0.025 & 0.0075 & 0.00175 \\ 0.0075 & 0.0070 & 0.00135 \\ 0.00175 & 0.00135 & 0.0043 \end{bmatrix}$$

where 0.025 is the variance of X_1 , 0.0075 is the covariance between X_1 and X_2 , 0.00175 is the covariance between X_1 and X_3 , and 0.0135 is the covariance between X_2 and X_3 .